JDBC

What is JDBC :-

- It is stand for **Java Database Connectivity.**
- Is an **API (Application Programming Interface)** in Java that allows Java applications to **connect, interact, and perform operations on databases** such as MySQL, PostgreSQL, Oracle, etc.

Definition :-

- JDBC is a **Java API used to connect** and **execute queries** on a database.

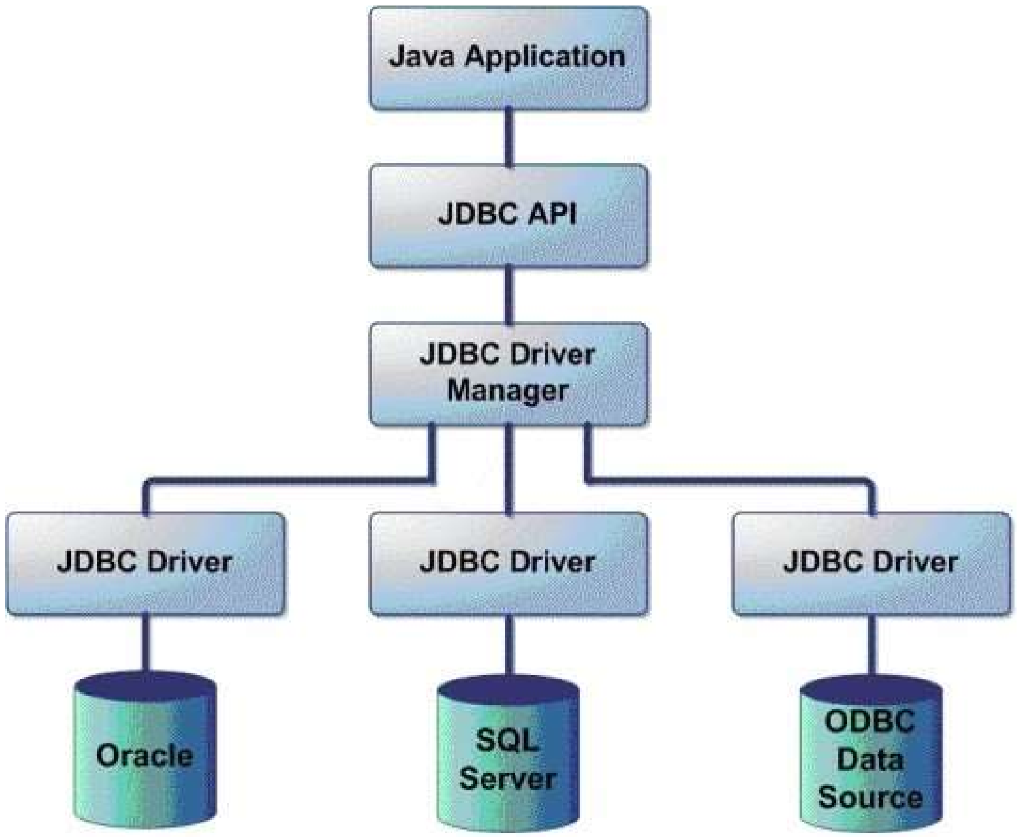
Purpose of JDBC :-

- To **connect Java applications** with any relational database.
- To **send SQL queries** and **retrieve results** from the database.
- To perform **CRUD operations** (Create, Read, Update, Delete).

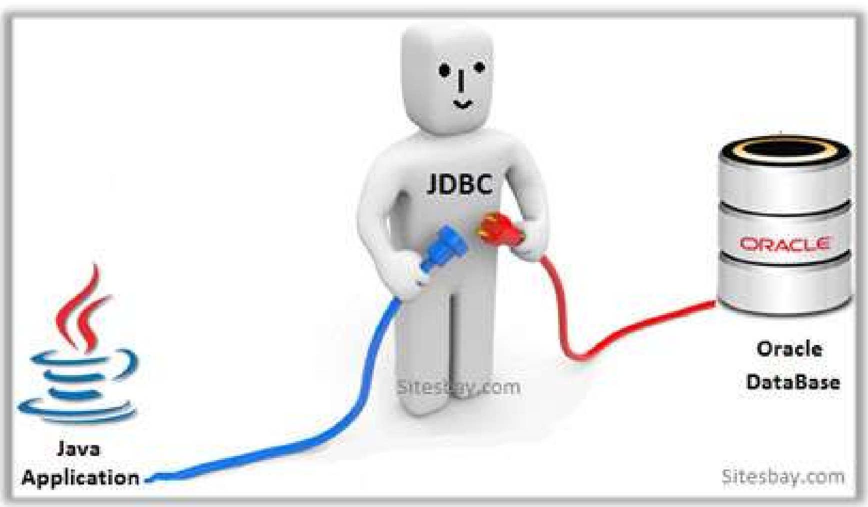
Architecture of JDBC:

1. **JDBC API :** Provides interfaces and classes for database access.
2. **JDBC Driver Manager :** Loads the correct driver for database connection.
3. **JDBC Driver :** Implements the API for specific databases (e.g., MySQL driver).
4. **Database :** Actual database like MySQL, Oracle, PostgreSQL, etc.

Architecture of JDBC :-



Working of JDBC :-



Implementation of JDBC :-

Input :-

```java
package JDBC;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.sql.Statement;

public class jdbc1 {

    public static void main(String[] args) throws ClassNotFoundException, SQLException {

        // 1.Load class of JDBC Driver

        Class.forName("org.postgresql.Driver");
        System.out.println("1.JDBC driver loaded...!!");

        // 2.Create Connection Portnumber, Username, Password

                // rdbmsdriver://localhost:portnumber/databasename
        String URL="jdbc:postgresql://localhost:5433/java8sept";
        String username="postgres";
        String password="Nikhil77";
        Connection con = DriverManager.getConnection(URL, username, password);
        System.out.println("2.Connection is created...!!");

        // 3.Create a Statement

        String create_table="create table studentdata(id int, name varchar(100))";
        Statement st = con.createStatement();
        System.out.println("3.Statement is created...!!");

        // 4.Execute the Statement

        st.executeUpdate(create_table);
        System.out.println("4.Query executed...!!");

        // 5.Close the Connection

        con.close();
        System.out.println("5.Connection close...!!");
    }
}
```

Output :-

```
1.JDBC driver loaded...!!
2.Connection is created...!!
3.Statement is created...!!
4.Query executed...!!
5.Connection close...!!
```

Process of JDBC :-